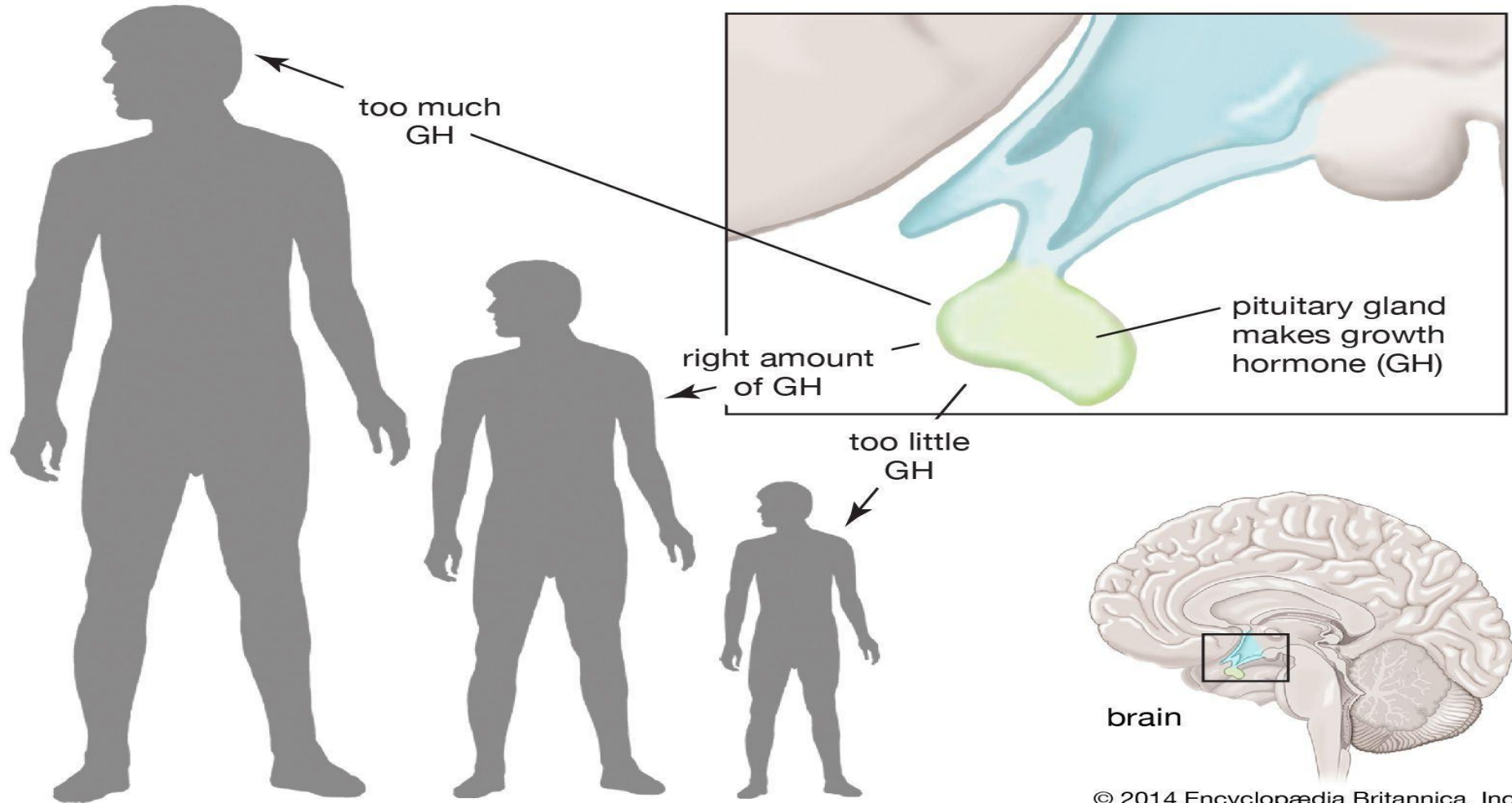


DISORDERS OF GROWTH HORMONES



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DR ABDUL SAMAD KHAN
SENIOR LECTURER -DIKIOS
DOW UNIVERSITY OF HEALTH SCIENCES

Objectives

Students should be able to

- Identify the factors that affect the growth beside the growth hormone.
- Describe the metabolic effect of growth hormone.
- Explain the functions of the growth hormone.
- Define the factors that influence the growth hormone secretion.
- Describe the growth hormone abnormalities.

GROWTH HORMONES

Growth hormone (GH), also known as human growth hormone (HGH) or somatotropin, is a peptide hormone produced by the pituitary gland that stimulates physical growth, cell regeneration, and metabolic regulation throughout life.

GH secretion is stimulated by growth hormone-releasing hormone (GHRH) and is inhibited by somatostatin.

The roles of growth hormone include influencing our height, and helping build our bones and muscles.

Natural levels of growth hormone fluctuate during the day, seemingly influenced by physical activity. For example, levels rise when we exercise.

Growth hormone levels increase during childhood and peak during puberty.

Growth hormone has both direct and indirect actions.

Direct actions include:

- Increased uptake of amino acids and production of proteins by the cells, increased breakdown of fats (lipolysis)
- release of fatty acids by the fat cells,
- increased synthesis and release of glucose (gluconeogenesis) into the blood by the liver,
- decreased uptake of glucose by muscle and fat cells.
- The overall direct effect of growth hormone is to increase lipid utilization and the promotion of protein synthesis and growth.

Indirect actions of growth hormone are mediated via messengers released by the liver.

- Growth hormone stimulates the secretion of a group of messengers called somatomedins, the most important of which are the insulin-like growth factors.
- These messengers stimulate the growth of cartilage and bone as well as increase protein synthesis in skeletal muscle.

Factors affect growth

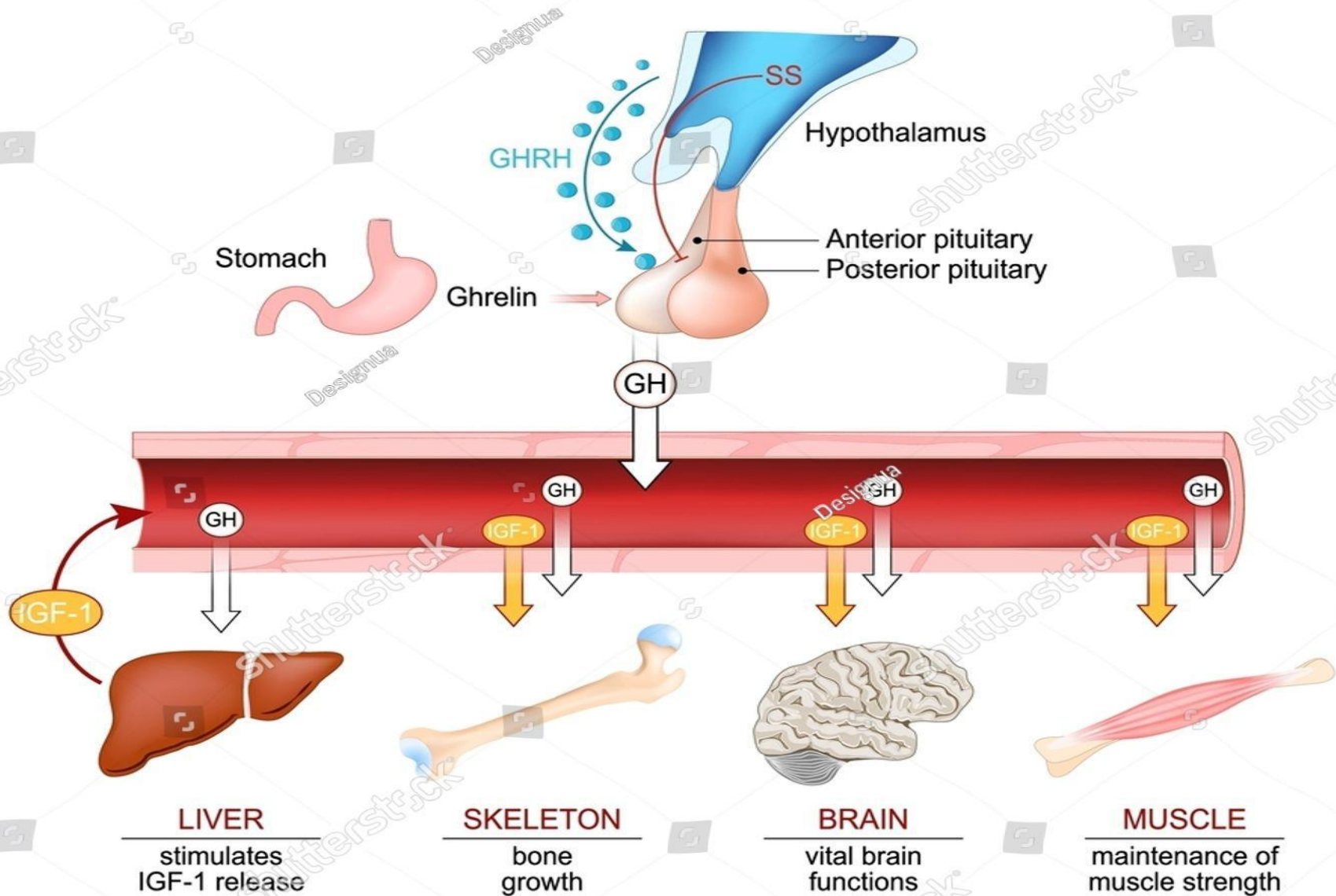
Growth depends on growth hormone but is influenced by other factors as well

- ☐ Genetic determination of an individual's maximum growth capacity.
- ☐ An adequate diet
- ☐ Freedom from chronic disease and stressful environmental conditions.
- ☐ Normal levels of growth-influencing hormones

HORMONES REGULATING GROWTH

- Thyroid Hormone
- Insulin: Stimulates IGF-1 (Somatomedin)
- Prolactin: Stimulates IGF-1
- Testosterone
- Estrogen

GROWTH HORMONE



GROWTH HORMONES SECRETION

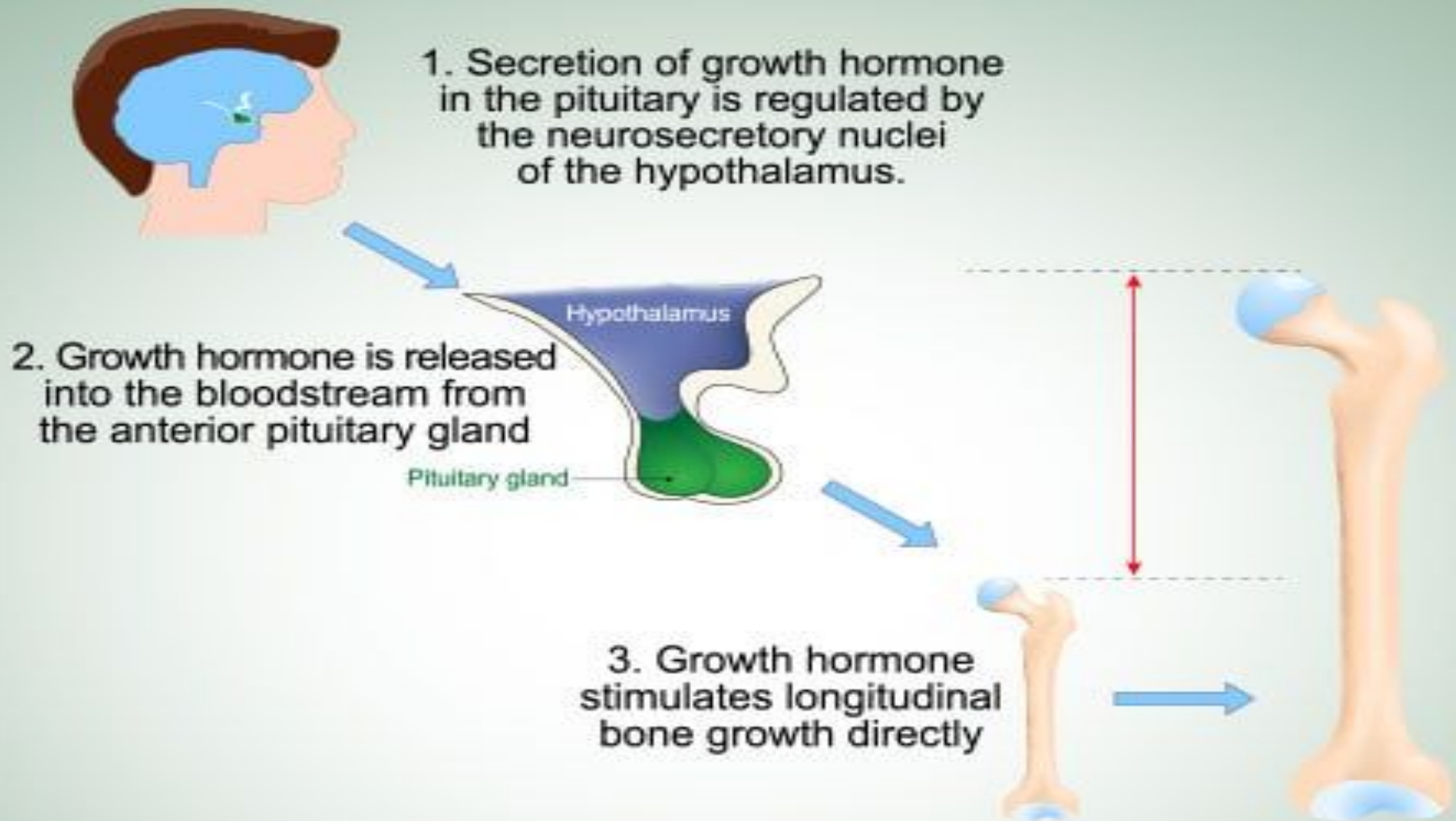
- Human growth hormone, also known as HGH and somatotropin, is a natural hormone secreted by pituitary gland that acts on many parts of the body to promote growth in children.
- Growth hormone is secreted by somatotropin cells which are the acidophilic cells of anterior pituitary
- Human growth hormone (HGH), also known as somatotropin, is a 191 amino acid single-chain polypeptide produced by somatotropin cells within the anterior pituitary gland.
- GH secretion is stimulated by growth hormone-releasing hormone (GHRH) and is inhibited by somatostatin.
- Secreted in bursts (not continuous)

- GH released mostly during sleep
 - optimal at night time
 - Changing sleeping pattern affects GH release
- GH production declines with age

Increased Growth Hormone Secretion:

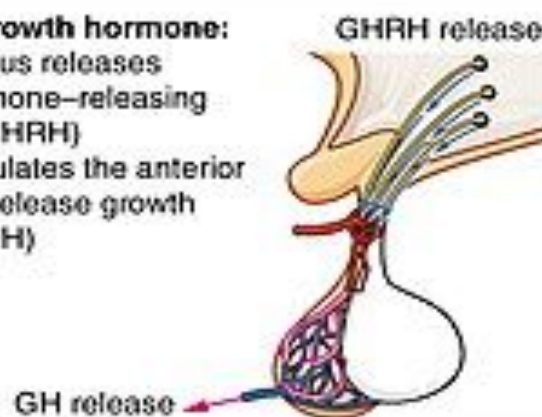
- Exercise regularly, 8 hours of sleep
- Protein-rich diet
- Avoid Stress

Mechanism of HGH



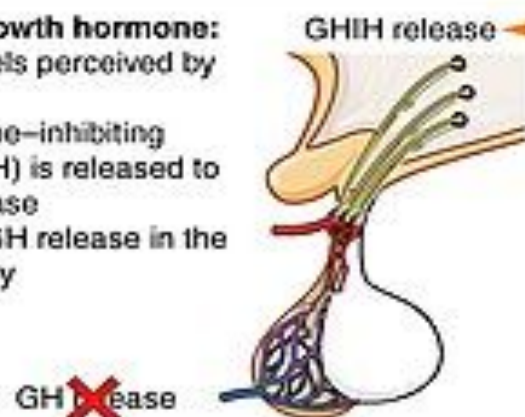
1) Release of growth hormone:

- Hypothalamus releases growth hormone–releasing hormone (GHRH)
- GHRH stimulates the anterior pituitary to release growth hormone (GH)



3) Inhibition of growth hormone:

- High IGF-1 levels perceived by hypothalamus
- Growth hormone–inhibiting hormone (GHIH) is released to inhibit GH release
- GHIH inhibits GH release in the anterior pituitary



2a) Glucose-sparing effect:

- Stimulates adipose cells to break down stored fat, fueling growth effects



2b) Growth effects:

- Increases uptake of amino acids from the blood
- Enhances cellular proliferation and reduces apoptosis

Targets:



Bone cells



Muscle cells



Nervous system cells



Immune system cells

2c) Diabetogenic effect:

- GH stimulates liver to break down glycogen into glucose, fueling growth effects



Liver releases IGF-1, further stimulating growth effects



Regulation of GH Secretion

Growth hormone secretion is altered by various factors. However, hypothalamus and feedback mechanism play an important role in the regulation of GH secretion

GH secretion is stimulated by:

1. Hypoglycemia
2. Fasting
3. Starvation
4. Exercise
5. Stress and trauma
6. Initial stages of sleep.

GH secretion is inhibited by:

1. Hyperglycemia
2. Increase in free fatty acids in blood
3. Later stages of sleep

GROWTH HORMONES ASSOCIATED DISORDERS

- Dwarfism
- Gigantism
- Acromegaly

DWARFISM

- ☐ Dwarfism is a medical condition defined by an adult height of 4 feet 10 inches (147 cm) or less, caused by genetic or medical factors.
- ☐ Deficiency in GH

Occurs in children

- GH absent during child's development
- Tumor causing hypopituitarism (decreased hormonal output)

HYPOACTIVITY OF ANTERIOR PITUITARY

1. Dwarfism

Dwarfism is a pituitary disorder in children, characterized by the stunted growth.

Causes

Reduction in GH secretion in infancy or early childhood causes dwarfism.

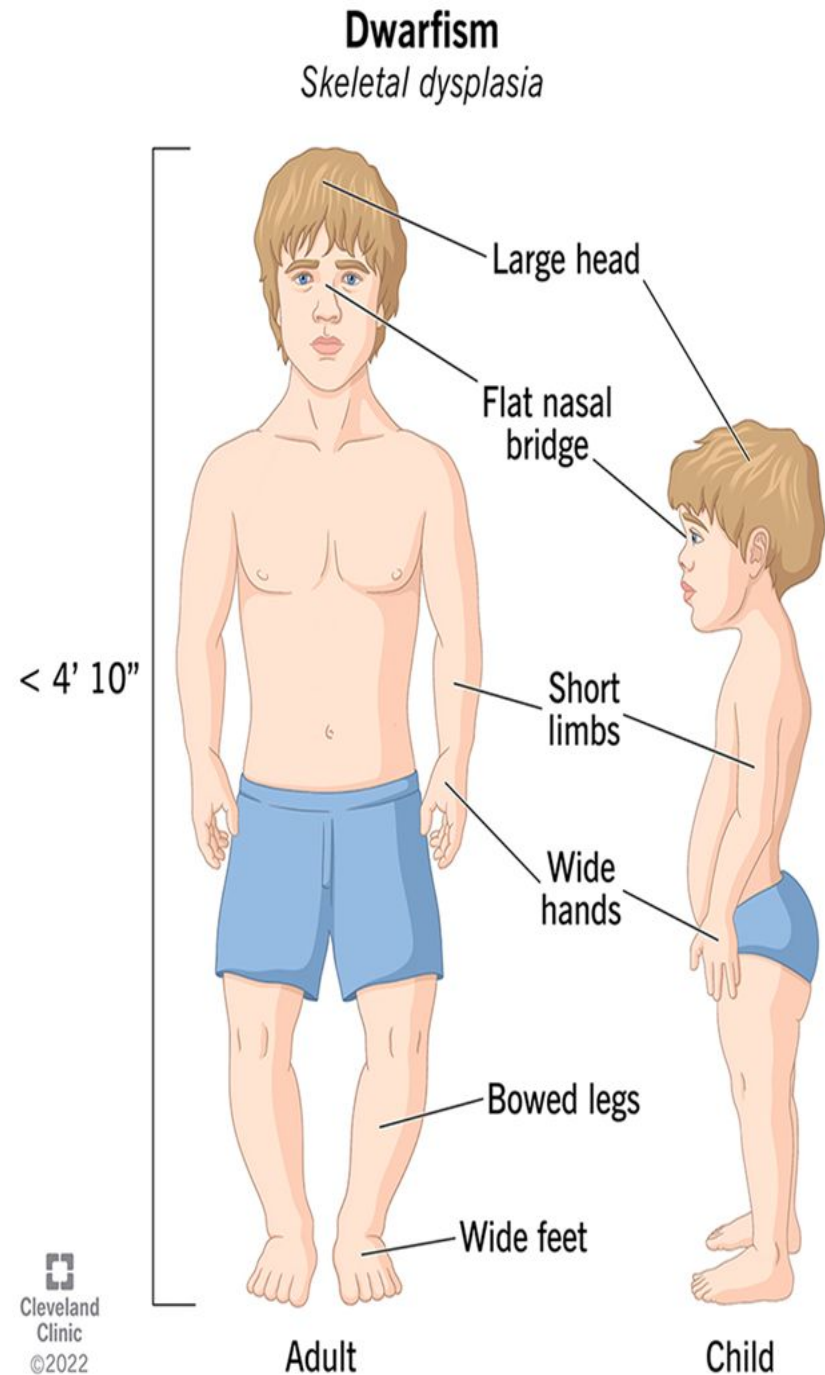
It occurs because of the following reasons:

I. Tumor of chromophobes: It is the most common cause for hyposecretion of GH, leading to dwarfism

II. Deficiency of GH-releasing hormone secreted by hypothalamus

III. Atrophy or degeneration of acidophilic cells in the anterior pituitary

IV. Panhypopituitarism: In this condition, there is reduction in the secretion of all the hormones of anterior pituitary gland.



Signs and symptoms

- A very short trunk.
- A short neck.
- Shortened arms and legs.
- Average-sized hands and feet.
- A broad, rounded chest.
- Slightly flattened cheekbones.
- An opening in the roof of the mouth, also called a cleft palate.
- Changes in hip structure that result in thighbones turning inward.
- A foot that's twisted or out of shape.
- Neck bones that are not stable.
- Hunching curvature of the upper spine that gets worse over time.
- Swayed lower back that gets worse over time.
- Vision and hearing problems.
- Arthritis and problems moving joints.
- Adult height ranging from 3 feet (91 cm) to just over 4 feet (122 cm).

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graph TD; A[DWARFISM SYMPTOMS] --> B[SHORT FINGERS]; A --> C[BOWING OF LEGS]; A --> D[SHORT ARMS AND LEGS]; A --> E[FLATTENED BRIDGE OF THE NOSE]; A --> F[LIMITED MOBILITY AT THE ELBOWS]; A --> G[ADULTS ARE ABOUT 4 FEET TALL]; A --> H[VISION AND HEARING PROBLEMS]; A --> I[A DISPROPORTIONATELY LARGE HEAD]; A --> J[ARTHRITIS]; A --> K[WIDE SPACES BETWEEN THE MIDDLE AND RING FINGERS]; A --> L[SWAYING OF THE BACK];
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DWARFISM SYMPTOMS

SHORT FINGERS

BOWING OF LEGS

SHORT ARMS AND LEGS

SWAYING OF THE BACK

WIDE SPACES BETWEEN THE MIDDLE AND RING FINGERS

FLATTENED BRIDGE OF THE NOSE

ARTHRITIS

VISION AND HEARING PROBLEMS

LIMITED MOBILITY AT THE ELBOWS

A DISPROPORTIONATELY LARGE HEAD

ADULTS ARE ABOUT 4 FEET TALL

Types of Dwarfism

Disproportionate Dwarfism

Some average-size parts of the body and some shorter-than-normal parts of the body



Achondroplasia is the most common disproportionate dwarfism, causing a normal-size torso but short limbs

Proportionate Dwarfism

The individual is smaller than average all over



Includes growth-hormone deficiency dwarfism, primordial dwarfism, and Seckel syndrome

2. Gigantism

Gigantism is the pituitary disorder characterized by excess growth of the body.

The subjects look like the giants with average height of about 7 to 8 feet.

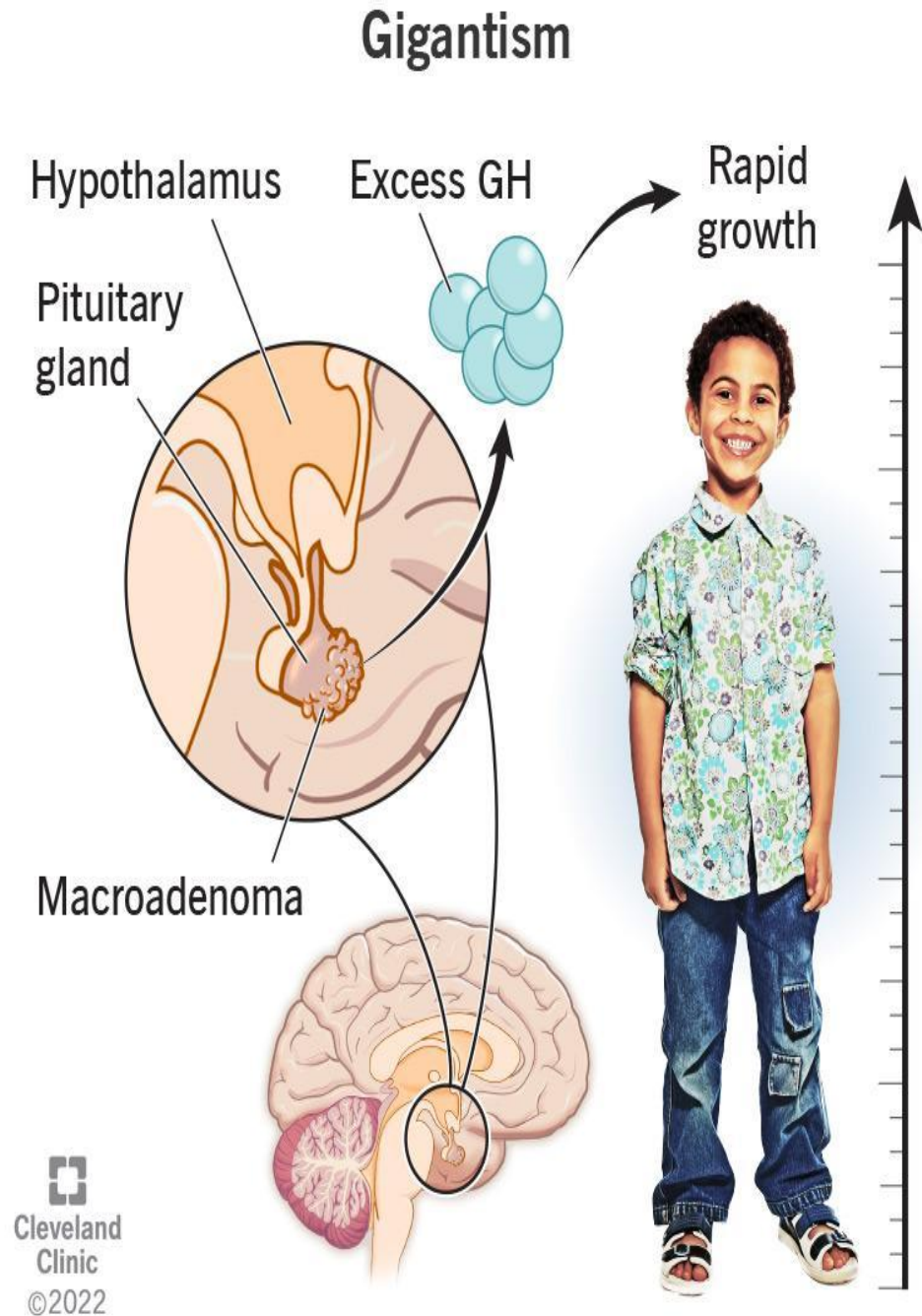
Causes:

Gigantism is due to hypersecretion of GH in childhood or in pre-adult life before the fusion of epiphysis of bone with shaft. Hypersecretion of GH is because of tumor of acidophil cells in the anterior pituitary.



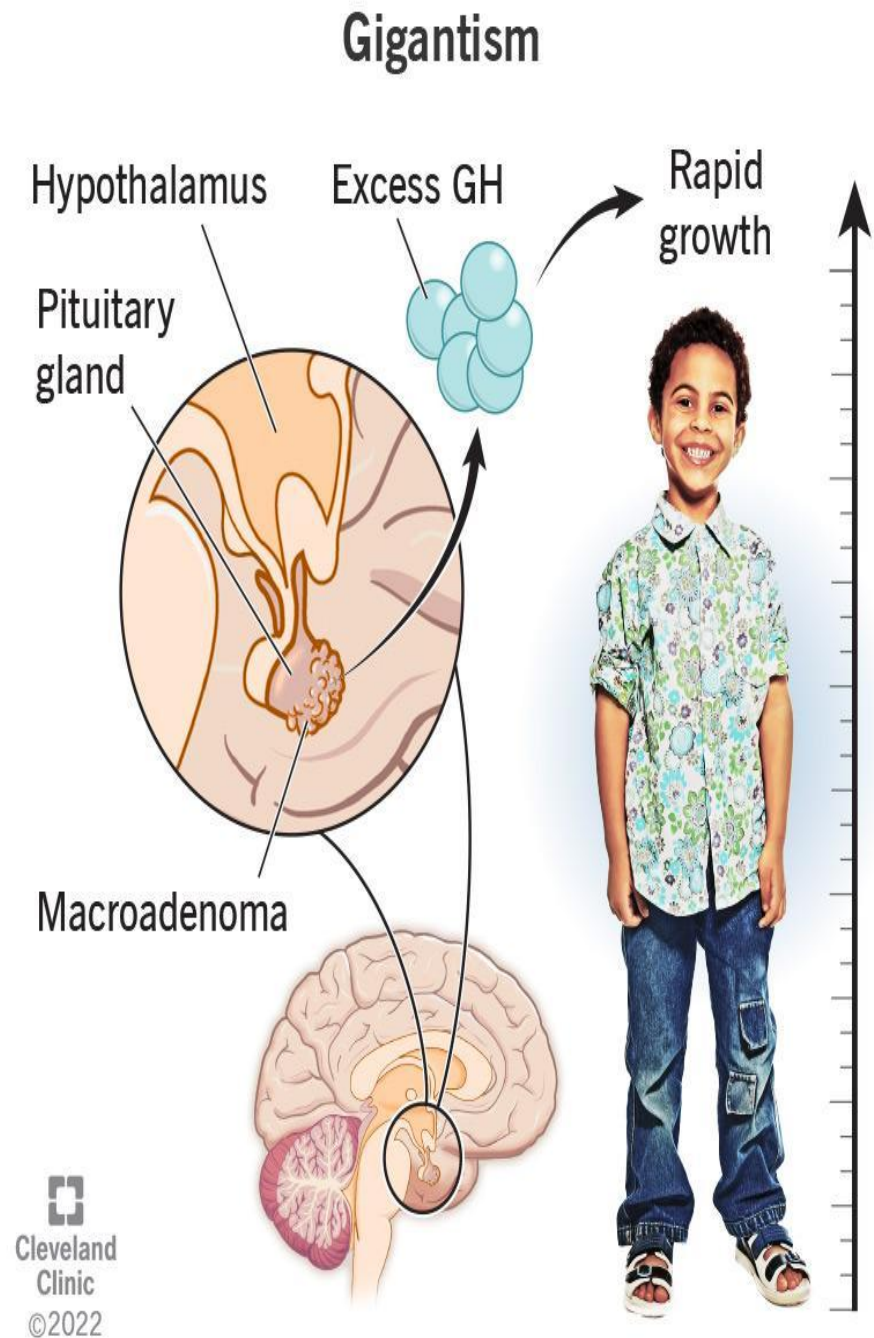
What is gigantism?

The pituitary gland normally produces GH, but a tumor on their pituitary can produce excess GH in gigantism.



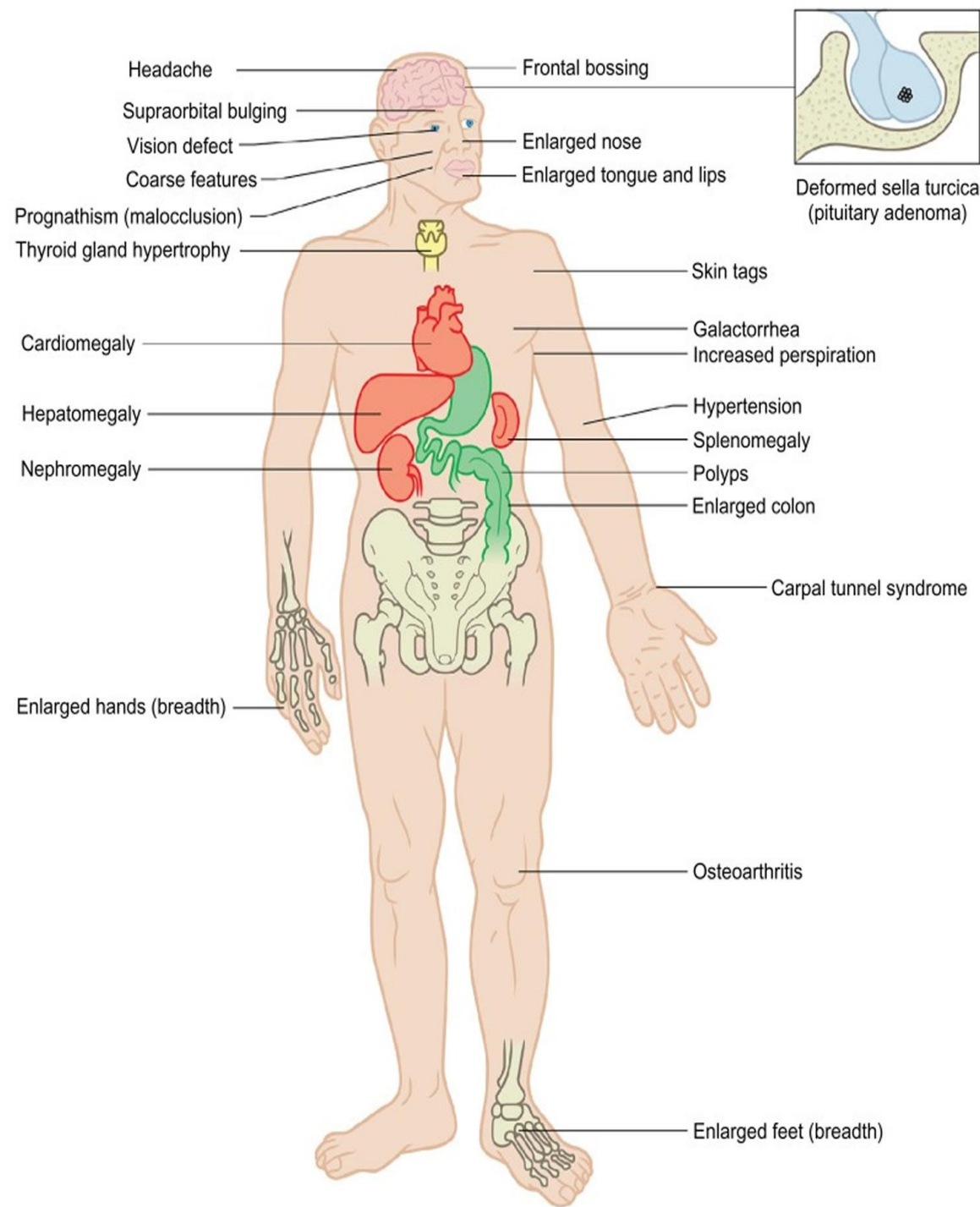
What is gigantism?

Gigantism, also called pediatric acromegaly and pituitary gigantism, is a very rare condition that happens when a child or adolescent has high levels of growth hormone (GH) in their body, which causes them to grow very tall. The pituitary gland normally produces GH, but a tumor on their pituitary can produce excess GH in gigantism.



Signs and symptoms of gigantism

- Unusually large hands and feet.
- Wide fingers and toes.
- Unusually prominent jaw and forehead.
- Coarse facial features (enlarged nose, lips and tongue)
- Gaps between the teeth.
- Excessive sweating.
- Joint pain.
- Muscle weakness.



Haji Mohammad Alam Channa (1953 – 2 July 1998) was a Pakistani farmer who was one of the tallest living people with the height of 7.8 feet (2.4 m). According to the Guinness Book of World Records, the tallest living man in the world (between 1982 and 1998), was Pakistan's Mohammad Alam Channa. Before his death in 1998, Channa was reported to be 7.7 feet (2.3 m) tall.



Robert Wadlow the world's Tallest Man (1918-1940)



8 feet 11 inches and 439 pounds when he died

GIGANTISM PROBLEMS

- Poor blood flow due to large body
- Increased muscle mass but weaker muscle
 - Excess GH produces salt in muscle tissues
 - Muscles swell with water
 - Results in disproportionate muscle growth 4 weaker muscles

Acromegaly

Acromegaly is the disorder characterized by the enlargement, thickening and broadening of bones, particularly in the extremities of the body.

Causes:

Acromegaly is due to hypersecretion of GH in adults after the fusion of epiphysis with shaft of the bone. Hypersecretion of GH is because of tumor of acidophil cells in the anterior pituitary.

Effects of Acromegaly:

Soft tissue harden

Deeper voice because larynx enlarges

Bigger tongue and lips

Cartilage in nose enlarges making nose broader

ACROMEGALY: PHYSICAL EFFECTS

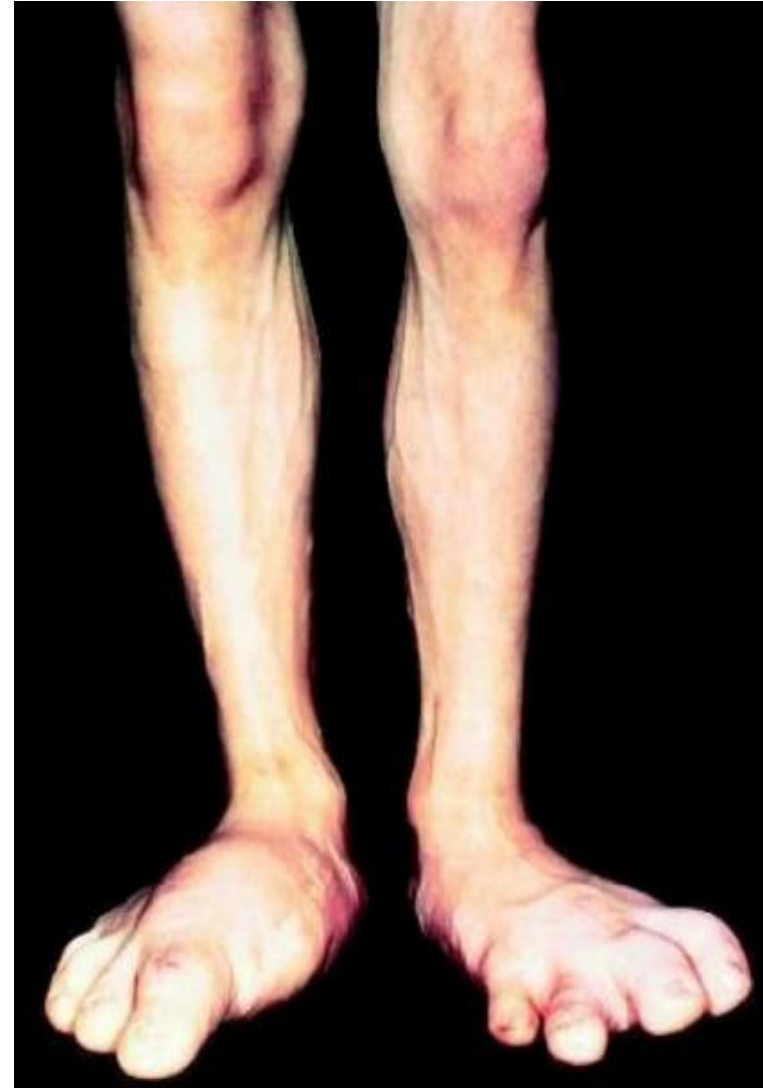


- Bone thickens
 - Forehead expands
 - Eyebrow ridges bulge outwards
 - Cheekbones more prominent
 - Mandible enlarges and pushes lower teeth outwards and become widely space



ACROMEGALY: EFFECTS ON MUSCLES DISORDERS

- Impaired Movements
 - Enlargement of bones crushes peroneal nerve in knee
 - Nerve carries messages to move foot and lower leg
 - Nerve cannot send messages to leg to trigger walking motion
- Also cause muscle numbness
- Leads to early death

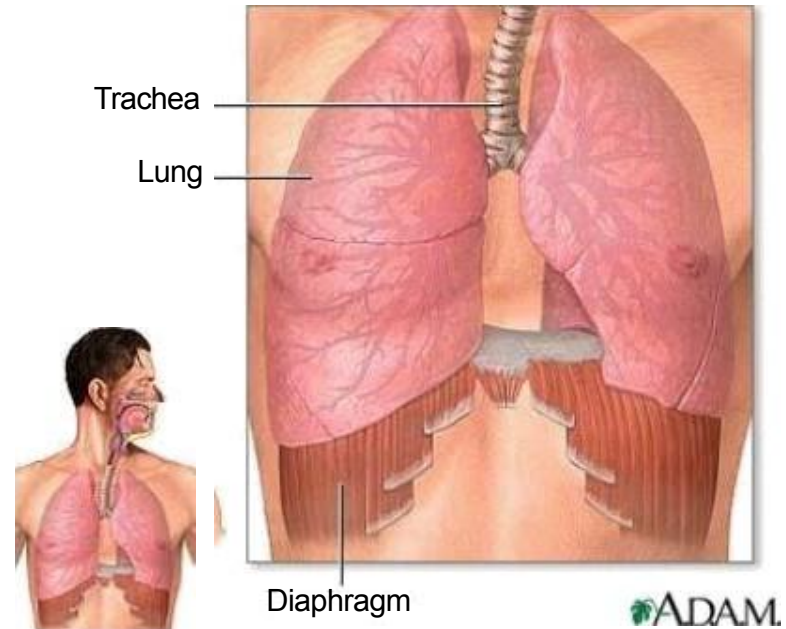


ACROMEGALY : HEART DEFECTS

- Heart tissue stiffen
 - heart cannot contract and relax
 - Ventricle harder to fill up
- Heart grows bigger in order to pump out sufficient blood

ACROMEGALY: LUNG DEFECTS ASSOCIATED DISORDERS

- As bones grow, rib cage expands
- Diaphragm is stretched thin and loses elasticity
- Breathing is reduced



Signs and Symptoms of Acromegaly

EYES - decreased vision
with large tumours

SKIN
-Thickening
-Oily
-Sweating
-Skin tags

HEART
- leaky valves
-Left ventricle thickening
(decreased pumping)

LUNGS
Rib changes inhibit
deep breathing.

LIVER/ Pancreas

-Makes glucose & IGF-1
-**insulin** Less effective
turning sugar into energy

Pre-cancerous gut polyps

Change in sodium
and water balance
from the kidneys can
result in high blood
pressure.

Poor Sexual function
Low testosterone
Irregular menses

BONES/ Muscles

-Enlarged hands and feet
-Carpel tunnel syndrome
-Joint changes, pain, arthritis
-Weakness



Acromegaly characteristic features

Prominent
supraorbital bridge

Teeth spacing
increased

Proximal
muscles weakness

Kyphosis

Hands are thick
and spade-like

Voice

Larynx

Prognathism
Hypertrophy

Adipose



What is the difference between gigantism and acromegaly?

Gigantism and acromegaly are both conditions that result from excess growth hormone (GH).

The difference is in who the conditions affect — adults develop acromegaly, whereas children and teenagers who are still growing develop gigantism.

In children, gigantism occurs when they experience excess GH before the growth plates in their bones fuse (before the end of puberty).

Once your growth plates have fused, excess GH causes acromegaly.

In this case, you don't grow in height, but the excess GH affects your bones' shape and your organ size as well as other health factors.

Gigantism is rarer than acromegaly.

The following chart compares gigantism and acromegaly:

	Gigantism	Acromegaly
Age of onset	childhood, before puberty	adulthood, after puberty
Cause	pituitary and non pituitary tumors	pituitary tumors or conditions like McCune-Albright syndrome, Carney complex, or GPR101 gene mutation
Main symptoms	tall height, large size, enlarged extremities and facial features	enlarged extremities and facial features
Onset of puberty	delayed	typical
Progress of symptoms	rapid	slow
Diagnosis	physical exam, imaging tests, blood tests	physical exam, imaging tests, blood tests
Preferred treatment	surgical removal of tumor	surgical removal of tumor